

Import Data

```
In [2]: import pandas as pd
```

```
In [23]: data = pd.read_csv(r"D:\Downloads\Udemy-PY\Housing+Data.csv")
```

Checking Metadata

```
In [7]: data.head()
```

```
Out[7]:
```

	date	area	average_price	code	houses_sold	no_of_crimes
0	1/1/1995	city of london	91449	E09000001	17.0	NaN
1	2/1/1995	city of london	82203	E09000001	7.0	NaN
2	3/1/1995	city of london	79121	E09000001	14.0	NaN
3	4/1/1995	city of london	77101	E09000001	7.0	NaN
4	5/1/1995	city of london	84409	E09000001	10.0	NaN

```
In [8]: data.count()
```

```
Out[8]: date          13549
area          13549
average_price  13549
code          13549
houses_sold   13455
no_of_crimes   7439
dtype: int64
```

```
In [9]: data.isnull().sum()
```

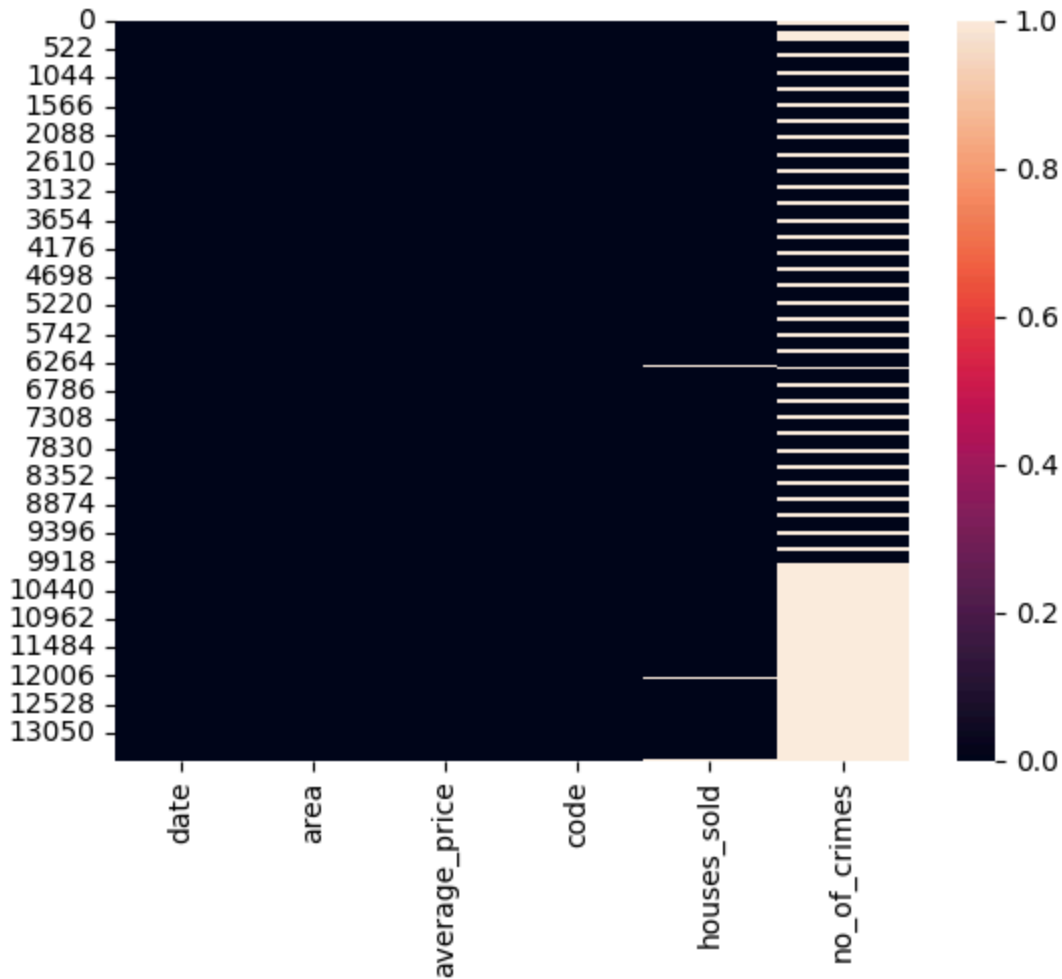
```
Out[9]: date          0
area          0
average_price  0
code          0
houses_sold    94
no_of_crimes  6110
dtype: int64
```

```
In [10]: import seaborn as sns
```

```
In [11]: import matplotlib.pyplot as plt
```

```
In [14]: sns.heatmap(data.isnull())
plt.show
```

```
Out[14]: <function matplotlib.pyplot.show(close=None, block=None)>
```



1. Convert datatype of Date into Date-Time format

In [25]: `data.dtypes`

```
Out[25]: date          object
area              object
average_price      int64
code              object
houses_sold        float64
no_of_crimes       float64
dtype: object
```

In [26]: `data.date = pd.to_datetime(data.date)`

2. Add a new column 'year' in the data frame which contains only year

In [27]: `data['year'] = data.date.dt.year`

In [28]: `data.head()`

Out[28]:

	date	area	average_price	code	houses_sold	no_of_crimes	year
0	1995-01-01	city of london	91449	E09000001	17.0	NaN	1995
1	1995-02-01	city of london	82203	E09000001	7.0	NaN	1995
2	1995-03-01	city of london	79121	E09000001	14.0	NaN	1995
3	1995-04-01	city of london	77101	E09000001	7.0	NaN	1995
4	1995-05-01	city of london	84409	E09000001	10.0	NaN	1995

3. Insert a new column of month at index 2

In [29]: `data.insert(1, 'month', data.date.dt.month)`In [30]: `data.head()`

Out[30]:

	date	month	area	average_price	code	houses_sold	no_of_crimes	year
0	1995-01-01	1	city of london	91449	E09000001	17.0	NaN	1995
1	1995-02-01	2	city of london	82203	E09000001	7.0	NaN	1995
2	1995-03-01	3	city of london	79121	E09000001	14.0	NaN	1995
3	1995-04-01	4	city of london	77101	E09000001	7.0	NaN	1995
4	1995-05-01	5	city of london	84409	E09000001	10.0	NaN	1995

4. Remove the columns 'month' and 'year'

In [33]: `data.drop(['month', 'year'], axis = 1, inplace = True)`In [35]: `data.head()`

Out[35]:

	date	area	average_price	code	houses_sold	no_of_crimes
0	1995-01-01	city of london	91449	E09000001	17.0	NaN
1	1995-02-01	city of london	82203	E09000001	7.0	NaN
2	1995-03-01	city of london	79121	E09000001	14.0	NaN
3	1995-04-01	city of london	77101	E09000001	7.0	NaN
4	1995-05-01	city of london	84409	E09000001	10.0	NaN

5. Show all those records where np. of crime is 0. How many such records are there?

In [39]: `len(data[data.no_of_crimes == 0])`

Out[39]: 104

6. What is the Minimum and Maximum 'average_price' per year in England?

In [45]: `data['year'] = data.date.dt.year`In [47]: `df1 = data[data.area == 'england']`In [48]: `df1.groupby('year').average_price.min()`

```
Out[48]: year
1995    52788
1996    52333
1997    55789
1998    61659
1999    65522
2000    75219
2001    84245
2002    96215
2003   121610
2004   139719
2005   158572
2006   166544
2007   181824
2008   165795
2009   159340
2010   174458
2011   173046
2012   174161
2013   176816
2014   188265
2015   202856
2016   220361
2017   231593
2018   240428
2019   243281
2020   247355
Name: average_price, dtype: int64
```

```
In [49]: df1.groupby('year').average_price.max()
```

```
Out[49]: year
1995      53901
1996      55755
1997      61564
1998      65743
1999      75071
2000      84191
2001      95992
2002     119982
2003     138985
2004     160330
2005     167244
2006     182031
2007     194764
2008     191750
2009     174136
2010     180807
2011     177335
2012     180129
2013     188544
2014     203639
2015     219582
2016     231922
2017     242628
2018     248620
2019     250410
2020     247355
Name: average_price, dtype: int64
```

7. What is the Average No. of crimes per area?

```
In [53]: data.groupby('area').no_of_crimes.mean()
```

```
Out[53]: area
barking and dagenham    1599.275109
barnet                  2278.441048
bexley                  1299.458515
brent                   2415.602620
bromley                 2020.131004
camden                  3056.572052
city of london          0.423423
croydon                 2652.943231
ealing                  2607.061135
east midlands           NaN
east of england         NaN
enfield                 2077.921397
england                 NaN
greenwich               2154.899563
hackney                 2575.751092
hammersmith and fulham  1914.676856
haringey                2406.427948
harrow                  1256.598253
haverling               1530.358079
hillington              2141.768559
hounslow                2040.231441
inner london           NaN
islington               2515.991266
kensington and chelsea  1833.222707
kingston upon thames    991.790393
lambeth                 3141.720524
lewisham                2296.458515
london                  NaN
merton                  1208.615721
newham                  2851.432314
north east             NaN
north west             NaN
outer london           NaN
redbridge               1940.480349
richmond upon thames    1041.620087
south east             NaN
south west             NaN
southwark               3028.563319
sutton                  1078.213974
tower hamlets           2542.253275
waltham forest          2113.598253
wandsworth              2209.716157
west midlands           NaN
westminster             5291.454148
yorks and the humber    NaN
Name: no_of_crimes, dtype: float64
```

8. Show those areas where Average price is less than 100000

```
In [59]: data[data.average_price < 100000 ].area.value_counts()
```

```
Out[59]: area
north east      112
north west      111
yorks and the humber  110
east midlands   96
west midlands   94
england         87
barking and dagenham  85
south west      78
east of england 76
newham          72
bexley          64
waltham forest  64
lewisham        62
havering        60
south east      59
greenwich       59
croydon         57
sutton          54
enfield         54
hackney         53
redbridge       52
southwark       48
tower hamlets   47
outer london    46
hillingdon      44
hounslow        41
lambeth         41
brent           40
london          39
merton          35
haringey        33
bromley         33
ealing          31
inner london    31
harrow          30
kingston upon thames 30
wandsworth      26
barnet          25
islington       19
city of london  11
Name: count, dtype: int64
```

```
In [ ]:
```